Serum phytoestrogens and prostate cancer risk in a nested case-control study among Japanese men

Kotaro Ozasa,[1] Masahiro Nakao,[2,6] Yoshiyuki Watanabe,[1] Kyohei Hayashi,[1] Tsuneharu Miki,[2] Kazuya Mikami,[2] Mitsuru Mori,[3] Fumio Sakauchi,[3] Masakazu Washio,[3] Yoshinori Ito,[4] Koji Suzuki,[4] Kenji Wakai[5,7] and Akiko Tamakoshi[5] for the JACC Study Group

Departments of [1]Epidemiology for Community Health and Medicine, [2]Urology, Kyoto Prefectural University of Medicine Graduate School of Medical Science, 465 Kajii-cho, Kamigyo-ku, Kyoto 602-8566; [3]Department of Public Health, Sapporo Medical University School of Medicine, S1 W17, Chuo-ku, Sapporo 060-8556; [4]Department of Public Health, Fujita Health University School of Health Sciences, 1-98 Dengakugakubo, Kutsukake-cho, Toyoake 470-1192; and [5]Department of Preventive Medicine/Biostatistics and Medical Decision Making, Nagoya University Graduate School of Medicine, 65 Tsurumai-cho, Showa-ku, Nagoya 466-8550

(Received September 2, 2003/Revised October 31, 2003/Accepted November 5, 2003)

The purpose of this study was to examine whether a high serum concentration of phytoestrogens reduces the risk of prostate cancer in a case-control study nested in a community-based cohort in Japan (Japan Collaborative Cohort (JACC) Study). Information on lifestyles and sera of the subjects were collected in 1988–90, and they were followed up to 1999. Incident and dead cases of prostate cancer and controls were matched for study area and age. Phytoestrogens and sex hormones in sera stored at −80°C were measured in 2002. Of 14,105 male subjects of the cohort who donated their sera, 52 cases and 151 controls were identified. Three datasets were analyzed; 1) all subjects, 2) 40 cases and 101 controls after excluding subjects with low testosterone levels who were suspected of having had medical intervention, and 3) 28 cases and 69 controls with prostate specific antigen level of ≤10.0 ng/ml. The odds ratio (OR) for the highest level to the lowest was 0.38 (95% confidence interval (CI); 0.13, 1.13) for genistein, 0.41 (0.15, 1.11) for daidzein, and 0.34 (0.11, 1.10) for equol for the second dataset. Genistein and daidzein showed similar findings in the third one. Equol and equol/daidzein ratio showed consistent findings in all three datasets (OR=0.39, 95% CI; 0.13, 0.89, trend *P*=0.02 for the first dataset). Their effects seemed to be independent of serum sex hormones. In conclusion, serum genistein, daidzein, and equol seemed to dose-dependently reduce prostate cancer risk. (Cancer Sci 2004; 95: 65–71)

Incidence and mortality rates of prostate cancer are much lower among Japanese men compared with Western populations.[1)] The incidence of prostate cancer in Japan ranges from around one-twentieth that of African-Americans to one-third that of some European populations, and the disparity in mortality is around one-ninth to one-third, respectively. Many other Asian countries also have lower incidences than Western countries.[2)]

Phytoestrogens, plant-origin isoflavonoids and lignans with estrogen-like activities, have been identified as key compounds that may decrease the risk of prostate cancer in Asian countries because Asian people consume more phytoestrogen-rich foods such as soybeans than Western people.[3,4)] The potential protective effect has been examined by epidemiological studies on the association of prostate cancer risk with isoflavonoid-rich diet or intake of isoflavonoids estimated from nutrient databases, and most of the studies support the effect, as summarized in a review.[4)] However, very few epidemiological studies have measured serum or urine phytoestrogens directly. A Nordic nested case-control study failed to show any protective effect of circulating enterolactone, a metabolite of lignans.[5)] A case-control study in Japan showed mostly unexpected results.[6)]

Epidemiologic data concerning blood levels of androgens are inconclusive,[7)] although a recent meta-analysis showed that a high level of total testosterone and low level of sex hormone-binding globulin (SHBG) increased the risk of prostate cancer when both of them were analyzed together.[8)] The effect of phytoestrogens may be confounded by endogenous sex hormones.

We therefore examined the protective effect of serum phytoestrogens on prostate cancer considering sex-hormone profiles in a case-control study nested in a community-based cohort in Japan.

Methods

Study population and serum samples. We carried out a nested case-control study as a part of the Japan Collaborative Cohort (JACC) Study, sponsored by Monbusho (the former Ministry of Education, Culture, Sports, Science and Technology of Japan).[9)] This study involved 48,019 male residents who were 40 years of age or older at baseline from 45 areas all over Japan. An epidemiological survey on lifestyle factors was conducted using a self-administered questionnaire from 1988 to 1990.

Those survey participants who underwent health-screening checks sponsored by municipalities were asked to donate blood samples during the same period as the questionnaire survey. Eventually, 14,105 men (29.3% of the respondents to the questionnaire survey) provided blood samples. Sera were separated from the samples at laboratories in or near the municipalities as soon as possible after the blood was drawn. The serum derived from each participant's sample was divided into three to five tubes (0.1–0.5 ml per tube), and the tubes were stored in deep freezers at−80°C until analyzed.

Informed consent was obtained from all participants. The Ethical Boards of the Nagoya University School of Medicine and the Kyoto Prefectural University of Medicine approved this study.

Case ascertainment and control selection. Subjects who had died or moved out of the study areas were identified using population registries, and their causes of death were confirmed from death certificates. Incident cases of cancer were identified by linkage with cancer registries in 24 study areas out of 45.[9)] Diagnosis of prostate cancer was defined by the code 'C61' in the International Statistical Classification of Diseases and Related Health Problems, 10th Revision.[10)] The subjects were followed up until the end of 1999, but the follow-up was censored earlier in some areas for incidence. Subjects who reported having had

E-mail: kozasa@basic.kpu-m.ac.jp
Present addresses: [6]Department of Urology, Meiji University of Oriental Medicine, Hiyoshi-cho, Funai-gun, Kyoto 629-0392; and [7]Division of Epidemiology and Prevention, Aichi Cancer Center Research Institute, 1-1 Kanokoden, Chikusa-ku, Nagoya 464-8681.

Abbreviations: JACC Study, Japan Collaborative Cohort Study; SHBG, sex hormone-binding globulin; BMI, body mass index; OR, odds ratio; CI, confidence interval; SD, standard deviation; CV, coefficients of variation; CMA, chlormadinone acetate; PSA, prostate specific antigen.

prostate cancer in the baseline survey were excluded from this study.

During the follow-up period, we identified 43 incident or dead cases of prostate cancer from 5711 subjects with baseline serum samples in the areas where a cancer registry was available, and 9 dead cases from 8394 in the other areas. These 52 cases were included in this study. We analyzed incident and dead cases together in order to maximize sample size for analysis. The mean time±SD from the baseline survey to the diagnosis of or death due to prostate cancer was 5.3±2.4 years. Three controls for each case were randomly selected from the subjects without death or any cancer incidence by matching for study area and age (as closely as possible). The control group was comprised of 151 males because five cases were matched to only two controls.

Biochemical assays of sera. Staff were blinded to the case-control status of samples, which were assayed at a single laboratory (SRL, Hachioji) in 2002. Serum levels of daidzein, genistein, and equol were measured by the method of Coward *et al*.[11] The sample was hydrolyzed prior to the measurement; serum was incubated with acetate buffer (pH 4.5) including β-glucuronidase/sulfatase (Nippon Biotest Laboratories, Inc., Kokubunji) at 37°C overnight. The dimethyl ether extract of the sample was dried under nitrogen flow, and redissolved in a solution of methanol, acetonitrile and water (2:1:3). The mixture was centrifuge-filtered using Ultrafree-MC, 0.22 μm pore size (Millipore, Bedford, MA), and the levels of daidzein, genistein, and equol were measured using an LC/MS/MS system (LC: HP1100 Series, Agilent Technologies, Palo Alto, CA, MS/MS: Quattro-Ultima, Micromass, Ltd., Manchester, UK). The ionizing method was electrospray using negative ions, and multiple reaction monitoring was used for mass analysis. The reference substances of genistein, daidzein, and equol were from Extrasynthese S.A. (Genay Cedex, France).

Total testosterone was measured by solid-phase radioimmunoassay (Coat-A-Count, Diagnostic Products Corporation (DPC), Los Angeles, CA), and SHBG was measured by immunoradiometric assay (IRMA-Count, DPC). Prostate specific antigen (PSA) was measured by solid-phase radioimmunoassay using Tandem-R PSA (Beckman Coulter, Inc., Fullerton, CA). The lower limits of detection were 0.5 ng/ml for phytoestrogens (1.9 n*M* for daidzein, 2.0 n*M* for genistein, and 2.1 n*M* for equol), 0.05 ng/ml (0.14 n*M*) for testosterone, 0.04 n*M* for SHBG, and 0.2 ng/ml for PSA. Total variation of measurements was examined for two samples for quality control in this study; coefficients of variation (CV) were 6.5% and 7.5% for genistein, 6.9% and 8.2% for daidzein, 8.2% and 9.1% for equol, and 5.6% and 6.9% for SHBG. CVs were around 5% to 12% for total testosterone, and around 2% to 5% for PSA, in routine measurements including the samples of this study.

Equol is a metabolite of daidzein, and is detected in sera or urine of some people ('equol-producers'), but not others ('non-equol-producers'). It has been found in general populations in both Western countries and Japan.[12–14] Equol-producers were defined as the subjects in whom serum equol was detected (≥1.9 n*M*) in this study.

Normal range of serum PSA concentration was assumed to be ≤4.0 ng/ml and serum PSA of >10.0 ng/ml is considered suggestive of clinical prostate cancer. This is because, first, the positive predictive value of prostate cancer for subjects with PSA of ≥10.0 ng/ml was 52.9% and that for subjects with PSA of 4.1–9.9 ng/ml was 26.1% among American volunteers,[15] and 44.1% and 8.4%, respectively, among participants in prostate cancer screening in Japan.[16] Second, serum PSA level is thought to correlate with the volume of prostate cancer,[17] and 59.5% of prostate cancers with PSA of ≥10.0 ng/ml involved seminal vesicles or lymph nodes while 16.4% of those with PSA of 2.8–10 ng/ml did so.[18]

Some hormonal treatments for prostate cancer or benign prostatic hypertrophy may modify the sex-hormone profile. Chlormadinone acetate (CMA), a steroidal androgen antagonist, has been widely used for the treatment of not only prostate cancer, but also benign prostatic hypertrophy in Japan. The total testosterone level was modified from around 5.0 to 1.5 ng/ml or lower on average by administration of 50 mg of CMA a day in a clinical trial.[19] The normal range of total testosterone in the Japanese male population is thought to be around 2.0 ng/ml or higher.[20] This limit is lower than that for Western people (3.0 ng/ml).[21] Therefore, subjects with suppressed testosterone level, i.e., <2.0 ng/ml (<6.9 n*M*), were assumed to have been hormonally treated at the time of blood donation.

The first analysis, including all subjects, evaluated the effect of genistein, daidzein, and equol, but not total testosterone and SHBG because the entire group included subjects with low levels of total testosterone in their sera (<6.9 n*M*). The second analysis was limited to subjects with serum total testosterone within normal range. Finally, the third analysis was further limited to cases with PSA≤10.0 ng/ml and controls with PSA<4 ng/ml. In addition, we analyzed the datasets in which cases diagnosed within the first 2 years of follow-up were excluded, together with the paired controls.

Statistical analysis. BMI [kg/m^2] was calculated from height and weight reported in the questionnaire at the baseline survey. The difference in means and proportions between cases and controls was examined by two-way analysis of variance and the χ^2 test. The cross-sectional relationships among age, BMI, serum daidzein, genistein, equol, total testosterone, and SHBG were examined using the Pearson correlation coefficient (using log-transformed values if necessary).

The risk of prostate cancer was evaluated in terms of odds ratios (ORs) and 95% confidence intervals (CIs) estimated in the conditional logistic model.[22] The ORs were computed according to tertiles of BMI, daidzein, genistein, total testosterone and SHBG among controls, and to three levels of equol and equol/daidzein ratio: nonequol-producers and bisection of equol-producers among controls. The ORs were adjusted by age because age was significantly different between cases and controls, but not by BMI due to similar values in cases and controls, as described in "Results." To test for linear trends in ORs over three levels, the median of each category was included in the logistic models.

To validate the procedures for collecting and storing the serum samples, differences in the sera of 151 controls between centers that collected and stored them were examined. The means of those serum components were examined for five centers with larger numbers of samples (12 to 36) and one group formed by combining all other centers with small numbers of samples (up to 6). Analysis was performed by one-way analysis of variance using log-transformed values.

Results

All subjects of 52 cases and 151 controls were included in the first analysis. Eleven cases and 27 controls with low serum levels of total testosterone (<6.9 n*M*) were excluded from the second analysis. Another one case and 23 controls were also excluded because they had no corresponding cases or controls with serum total testosterone at the normal level. Therefore, 40 cases and 101 controls were available for the second analysis. Twenty-five cases had three controls for each, 11 had two controls, and four had one control. In the third analysis, 28 cases and 69 controls were available because 12 cases showed high levels of PSA (>10.0 ng/ml). The highest PSA level was 8.6 ng/ml in the cases, and 3.7 ng/ml in the controls. Sixteen cases had three controls for each, nine had two controls, and three had one control in this dataset.

Characteristics of these cases and controls are shown in Table 1. Cases were significantly older than controls (P=0.0001). The BMI level was almost the same between cases and controls. Serum genistein and daidzein for all subjects, total testosterone and SHBG for the subjects with total testosterone at the normal level, and equol for equol-producers were log-normally distributed. Serum equol was significantly higher in controls than in cases among equol-producers, but the proportion of equol-producers did not differ (67.3% for cases and 75.3% for controls, P=0.38). Serum genistein and daidzein did not differ between cases and controls. Serum total testosterone and SHBG did not differ between cases and controls in the dataset for the second analysis, and other characteristics were similar to the first one. Characteristics of the dataset used for the third analysis were also similar to the first and second ones.

Table 2 shows the relationship between age, BMI, serum phytoestrogens, and sex hormones among the analyzed controls. Age was negatively associated with BMI, and positively with SHBG. BMI was negatively associated with total testosterone and SHBG. Serum genistein and daidzein were very strongly correlated with each other, and were moderately correlated with serum equol. Serum total testosterone and SHBG were also moderately associated with each other. The results among controls for the second and third analyses were substantially the same as those for the first analysis.

Age-adjusted ORs and 95% CIs for serum phytoestrogens and sex hormones by conditional logistic regression analysis are shown in Tables 3–5 for the first to third analyses, respectively. BMI was not associated with prostate cancer risk in any analysis (data not shown in Tables). Genistein and daidzein showed a tendency toward dose-dependent risk reduction of prostate cancer in the second and third analyses (Tables 4, 5). The ORs were around 0.4–0.5 for the highest serum level to the lowest. Genistein and daidzein showed unclear results in the first analysis (Table 3).

Equol and equol/daidzein ratios showed a tendency toward dose-dependent risk reduction of prostate cancer in all analyses (Tables 3–5). The risk was around one-third for the upper half level of equol-producers compared with nonequol-producers. Equol-producers as a whole also tended toward half the level of risk of nonequol-producers. These results showed statistical significance for equol and equol/daidzein ratio in the first analysis (Table 3), and marginal significance for most phytoestrogens in the second analysis (Table 4). Serum SHBG and total testosterone were analyzed together in the second and third analyses. SHBG correlated with a reduced risk of prostate cancer in the second and third analyses (Tables 4, 5). When ORs for phytoestrogens were adjusted by age, total testosterone, and SHBG in the second and third analyses, they were very similar to the ORs adjusted only by age (data not shown).

Three cases were diagnosed within the first 2 years of follow-up. Two of these three were part of the second dataset, and one was in the third dataset. The results from these datasets in which these cases were excluded were similar to the results shown above (data not shown).

The means for genistein and daidzein in all controls significantly differed among the five centers and one group of combined centers (P=0.039 and 0.010, respectively), but those for equol and SHBG did not (P=0.26 and 0.12, respectively). The mean for testosterone was examined among controls with a normal range of testosterone, and did not differ among centers (P=0.45). The mean PSA value was examined among centers after excluding both abnormally elevated samples (>4 ng/ml) and those showing suppression, probably owing to medical treatment (<0.2 ng/ml), and these values did not differ either (P=0.95).

Table 1. Characteristics of cases and controls

		Cases			Controls			P[2]
		N	Mean[1]	25 and 75 percentiles	N	Mean[1]	25 and 75 percentiles	
All subjects								
Age	(years)	52	69.4	64, 74	151	68.7	63, 73	0.0001
BMI	(kg/m^2)	49	22.3	20.5, 24.7	140	22.3	20.2, 24.4	0.95
Genistein	(nM)	52	321.5	195.4, 647.2	151	367.9	184.6, 849.6	0.44
Daidzein	(nM)	52	122.0	72.1, 244.5	151	139.0	64.1, 320.1	0.46
Equol (for equol producers)	(nM)	35	33.6	14.0, 78.0	113	55.40	28.6, 145.9	0.041
Subjects for the second analysis[3]								
Total testosterone	(nM)	40	16.1	13.2, 19.5	101	15.7	12.6, 19.5	0.61
SHBG	(nM)	40	40.4	29.7, 53.9	101	43.7	34.5, 59.0	0.20

1) Geometric means for genistein to SHBG.
2) Difference between cases and controls, tested by two-way analysis of variance or chi-square test.
3) Subjects with normal total serum concentration of total testosterone.

Table 2. Pearson's correlation coefficients between age, BMI and serum phytoestrogens and sex hormones

		Age	BMI	Genistein	Daidzein	Equol	Total testosterone
All controls	BMI	−0.18*					
	Genistein[1]	0.06	0.07				
	Daidzein[1]	0.09	0.10	0.91**			
	Equol[1, 2]	−0.07	−0.12	0.51**	0.40**		
Controls for the second analysis	Total testosterone[1]	0.07	−0.17†	0.04	−0.02	0.09	
	SHBG[1]	0.36**	−0.30**	0.14	0.09	0.03	0.54**

† P<0.1, * P<0.05, ** P<0.01.
1) Calculated using log-transformed values.
2) Limited to equol-producers.

Discussion

A protective effect of phytoestrogens on prostate cancer was predicted from ecological observations,[1] and has been examined in studies on the association of prostate cancer risk with isoflavonoid-rich diet or intake of isoflavonoids estimated using nutrient databases.[4] However, no reports have shown a direct inverse association between serum or urine phytoestrogen levels and prostate cancer risk.[5] This study showed that a high serum phytoestrogen concentration tended to reduce the risk of prostate cancer in a nested case-control study, part of a community-based cohort study in Japan. Genistein, daidzein, and equol all showed a probable protective effect. Of the three, equol showed the most consistent results through three analyses that used variously defined subjects. The findings for genistein and daidzein were rather obscure in the analysis for all subjects, although they showed a tendency toward dose-dependent risk reduction in the second and third analyses that excluded the subjects with low serum total testosterone levels. Dietary habits of subjects with low testosterone levels might have been biased, as they could have been hormonally treated and thus may have changed their diets as well.

Equol has been detected at high levels in urine, plasma, or feces in some populations (equol-producers), and at almost null levels in others (nonequol-producers).[12–14] Isoflavones are contained in soy mostly as glycosides, and are hydrolyzed to the aglycones after ingestion. They can be absorbed or metabolized further by intestinal microflora before absorption.[13, 14] Equol

Table 3. Age-adjusted odds ratios (ORs) and 95% confidence intervals (CIs) of prostate cancer for serum phytoestrogens and sex hormones by conditional logistic regression analysis among all subjects (first analysis)

		Cases	Controls	OR	95% CI	*P*	Trend *P*
Genistein (n*M*)	<239	18	51	1.00			
	239–682	22	50	1.37	0.61, 3.07	0.43	
	>682	12	50	0.76	0.32, 1.82	0.54	0.46
Daidzein (n*M*)	<89	18	51	1.00			
	89–239	21	50	1.43	0.63, 3.25	0.38	
	>239	13	50	0.74	0.31, 1.76	0.50	0.38
Equol (n*M*)	<1.9[1)]	17	38	1.00			
	1.9–56.1	24	57	0.70	0.31, 1.56	0.38	
	>56.1	11	56	0.39	0.15, 0.98	0.046	0.053
Equol/daidzein ratio	0[1)]	17	38	1.00			
	0.01–0.44	24	57	0.80	0.35, 1.86	0.61	
	>0.44	11	56	0.35	0.13, 0.89	0.028	0.023
Nonequol-producers		17	38	1.00			
Equol-producers		35	113	0.55	0.26, 1.16	0.11	

1) Not detected.

Table 4. Age-adjusted odds ratios (ORs) and 95% confidence intervals (CIs) of prostate cancer for serum phytoestrogens and sex hormones by conditional logistic regression analysis among the subjects with serum total teststerone of ≥6.9 n*M* (second analysis)

		Cases	Controls	OR	95% CI	*P*	Trend *P*
Genistein (n*M*)	<332	19	33	1.00			
	332–765	14	35	0.85	0.36, 1.98	0.71	
	>765	7	33	0.38	0.13, 1.13	0.084	0.085
Daidzein (n*M*)	<126	22	33	1.00			
	126–320	10	35	0.46	0.17, 1.23	0.11	
	>320	8	33	0.41	0.15, 1.11	0.080	0.098
Equol (n*M*)	<1.9[2)]	14	23	1.00			
	1.9–60.7	18	39	0.73	0.27, 1.90	0.52	
	>60.7	8	39	0.34	0.11, 1.10	0.074	0.072
Equol/daidzein ratio	0[2)]	14	23	1.00			
	0.01–0.42	17	39	0.88	0.30, 2.54	0.82	
	>0.42	9	39	0.34	0.11, 1.07	0.067	0.052
Nonequol-producers		14	23	1.00			
Equol-producers		26	78	0.56	0.23, 1.37	0.20	
Total testosterone[1)] (n*M*)	6.9–13.9	11	33	1.00			
	14.0–18.7	17	35	1.40	0.50, 3.96	0.51	
	>18.7	12	33	1.22	0.36, 4.14	0.74	0.77
SHBG[1)] (n*M*)	<37.3	16	33	1.00			
	37.3–54.8	15	35	0.88	0.32, 2.37	0.80	
	>54.8	9	33	0.42	0.12, 1.49	0.18	0.13

1) Total testosterone and SHBG were mutually adjusted.
2) Not detected.

Table 5. Age-adjusted odds ratios (ORs) and 95% confidence intervals (CIs) of prostate cancer for serum phytoestrogens and sex hormones by conditional logistic regression analysis among the subjects with serum total testosterone of ≥6.9 n*M* and PSA ≤10 ng/ml for cases and <4ng/ml for controls[3] (third analysis)

		Cases	Controls	OR	95% CI	*P*	Trend *P*
Genistein (n*M*)	<300	12	23	1.00			
	300–781	10	23	0.90	0.34, 2.40	0.83	
	>781	6	23	0.42	0.12, 1.38	0.15	0.15
Daidzein (n*M*)	<127	13	23	1.00			
	127–317	8	23	0.61	0.21, 1.76	0.36	
	>317	7	23	0.51	0.17, 1.52	0.22	0.25
Equol (n*M*)	<1.9[2]	10	14	1.00			
	1.9–57.0	11	27	0.63	0.18, 2.14	0.46	
	>57.0	7	28	0.33	0.08, 1.31	0.11	0.12
Equol/daidzein ratio	0[2]	10	14	1.00			
	0.01–0.38	11	27	0.67	0.19, 2.30	0.52	
	>0.38	7	28	0.30	0.07, 1.29	0.10	0.10
Nonequol-producers		10	14	1.00			
Equol-producers		18	55	0.49	0.16, 1.55	0.22	
Total testosterone[1] (n*M*)	6.9–13.9	10	23	1.00			
	14.0–19.1	11	23	0.92	0.29, 2.86	0.88	
	>19.1	7	23	0.61	0.18, 2.10	0.44	0.45
SHBG[1] (n*M*)	<37.9	14	23	1.00			
	37.9–53.6	9	23	0.62	0.20, 1.94	0.41	
	>53.6	5	23	0.27	0.07, 1.05	0.059	0.059

1) Total testosterone and SHBG were mutually adjusted.
2) Not detected.
3) The highest PSA level was 8.6 ng/ml in cases and 3.7 ng/ml in controls.

excretion in urine was not associated with soy intake, although urinary genistein, daidzein, and the sum of isoflavonoids were well correlated with soy intake.[13, 23] In our study, serum equol level was moderately correlated with serum daidzein level (r=0.40). Serum daidzein level was not different between equol-producers and nonequol-producers (geometric means; 142 nM vs. 137 nM, P=0.90 for all controls). Equol/daidzein ratios seemed to reflect metabolic activity from daidzein to equol. The ORs of the ratio were similar to those of equol concentration. Thus, the protective effect on prostate cancer may be directly due to equol concentration.

A case-control study for examining the effect of phytoestrogens on prostate cancer was conducted by Akaza *et al*. in Japan.[6] Serum levels of genistein, daidzein, and equol were all higher in cases than in controls although the proportion of equol-producers was lower in cases (39.7%) than in controls (50.0%) (P=0.10). These unexpected results may be caused by the fact that case patients had been diagnosed as having prostate cancer at the time of collecting sera, and their diet might have been changed. Unknown factors relating to the ability to produce equol might be involved in reducing the risk of prostate cancer, in view of the difference in the proportion of equol-producers between the cases and controls.[6]

Prostate cancer has a long natural history before becoming clinically detectable. In the second analysis in this study, subjects with low serum total testosterone were excluded because they might have been hormonally treated for prostate cancer or prostate hypertrophy at the baseline survey. In the third analysis, the subjects were limited by PSA concentration, and the highest PSA level was 8.6 ng/ml among the cases and 3.7 ng/ml among the controls. The results suggested that there were few clinical and undiagnosed prostate cancers among the cases and even fewer in the controls. We also analyzed the modified datasets in which the cases diagnosed within the first 2 years were excluded. The results of these analyses were consistent with the corresponding original analyses.

Subjects who were aware of their prostate disease may have changed their lifestyles. Disease history was requested in the baseline questionnaire, but the question was not focused on prostate diseases. However, subjects probably would not have changed their lifestyles if they were not aware of the disease. There is some argument about the biological effect of prostate cancer on sex-hormone profiles,[24] but the effects of phytoestrogens were mostly independent of sex hormones in this study. Therefore, the result of the second analysis (Table 4) was considered valid because it was consistent with the third one (Table 5).

A high level of SHBG tended to reduce the risk of prostate cancer in this study. This finding is consistent with a recent meta-analysis,[8] although the role of sex hormones in prostate cancer is inconclusive.[7] Phytoestrogens were thought to reduce the prostate cancer risk independently of sex hormones in this study. In Japanese men, it was reported that intake of soy product was inversely associated with serum estradiol and total and free testosterone (Spearman r=−0.32 (P=0.009),−0.25 (P=0.05), and−0.25 (P=0.06), respectively), but not with SHBG (r=−0.08).[25] Soymilk supplementation also might decrease the circulating estrone concentration, but did not alter the total and free testosterone, or SHBG concentration in Japanese men.[26] In this study, virtually no correlation was observed between serum phytoestrogens and sex hormones.

The mechanism of phytoestrogens in reducing the risk of prostate cancer seems to involve the role of estrogen receptor β in prostate tissue, although other mechanisms including inhibition of serum prostate-specific antigen, induction of anti-carcinogenic gene expression, and inhibition of tumor-related cytokines are possible.[3, 4] Estrogen activity of isoflavones has been evaluated by various methods. In assay of estrogen receptor-dependent β-galactosidase induction in yeast cells, equol showed the strongest potency (several times higher than genistein and many times higher than daidzein).[27–29] In proliferation assay of estrogen-sensitive human cancer cell lines,

genistein was determined to be the strongest in two reports,[28, 30] and equol was the strongest in one.[29] By competition binding assays with human estrogen receptors, genistein had the strongest potency, followed by equol and daidzein.[29] Recently, it was reported that equol strongly inhibited growth of benign and malignant prostatic epithelial cells *in vitro* at concentrations that can be obtained naturally through dietary soy consumption.[31] Modification of serum estrogen and androgen levels by phytoestrogens should still be considered; this includes inverse associations between intake of soy products and sex hormones,[26] and a favorable sex-hormone profile in equol-producers.[32]

The pharmacokinetics of phytoestrogens in plasma has been investigated in Japanese populations.[12, 33] Watanabe *et al.*[12] showed that the plasma concentration of both genistein and daidzein reached their highest values around 6 h after single ingestion of baked soybean powder. The peak of equol was later (8–12 h). The half-lives of plasma genistein and daidzein were 8.36 and 5.79 h, respectively. Izumi *et al.*[33] found a rather faster turn-over time for the aglycone form of isoflavones in plasma. The concentrations of phytoestrogens in our subjects were consistent with those in the previous reports on Japanese men,[34, 35] despite the difference in measurement methods and the long-term storage of our samples.

Serum concentrations of phytoestrogens were compared with the intake of soybean products in our study. At the baseline survey of this cohort study, the subjects indicated their average consumption of tofu and miso-soup during the previous 1 year. Simple regression coefficients for tofu intake among the 151 controls overall were 0.34 (P=0.0010) for serum genistein concentration and 0.35 (P=0.0006) for serum daidzein concentration (log-transformed values in mol/liter), when the values for the independent variable (tofu) were assigned as 1=scarcely any tofu intake, 2=1–2 times/month, 3=1–2/week, 4=3–4/week, and 5=almost every day. In the case of miso-soup, we obtained simple regression coefficients of 0.14 (P=0.12) for genistein and 0.19 (P=0.046) for daidzein (independent variable; 1=every other day or less, 2=one bowl of miso-soup/day, 3=2/day, 4=3/day, and 5=4+/day). For equol, however, the regression coefficients were 0.11 (P=0.38) in the case of tofu and −0.05 (P=0.65) in the case of miso-soup among equol-producers. Serum genistein and daidzein were associated with available dietary information, but equol was not.

These associations between information on long-term intake of soybean products and serum concentrations of genistein and daidzein in one-point samples, despite the short serum half-lives of these compounds, showed the stability of a lifestyle involving consumption of soybean products. Frequent intake of phytoestrogen-rich foods may be required to maintain blood concentration levels. An unusually large intake of these foods before blood donation may have produced an incidental high blood concentration of phytoestrogens, but this seems unlikely to have had a significant effect because of the apparent lifestyle stability. In contrast, serum equol concentrations seemed to be less affected by dietary habits, possibly because serum equol concentration depends on the metabolism of daidzein. Serum testosterone levels may fluctuate in a subject, but adjusting for testosterone did not alter the association of phytoestrogens with prostate cancer.

The difference in means of serum genistein and daidzein between centers may reflect regional differences in the consumption of soybean products such as tofu and natto in Japan. Error in handling samples is not a plausible explanation because intrinsic components such as equol, testosterone, SHBG, and PSA did not differ significantly.

ORs in this study were adjusted by age because cases were significantly older than controls, despite matching. Although the differences were small, most cases were older than controls. We did not use BMI for adjustment in this study although it was empirically used in other studies.[8] BMI did not differ significantly between cases and controls, and the ORs adjusted by BMI were similar to the unadjusted ones (data not shown). Family history of prostate cancer could not be evaluated, because only one case reported it.

In conclusion, high serum levels of phytoestrogens, especially equol, seemed to reduce prostate cancer risk, and their effects seemed to be independent of serum total testosterone and SHBG levels.

The authors wish to express their sincere appreciation to Dr. Kunio Aoki, Professor Emeritus, Nagoya University School of Medicine and the former chairman of the JACC Study Group, and also to Dr. Haruo Sugano, the former Director of the Cancer Institute of the Japanese Foundation for Cancer Research, who greatly contributed to the initiation of the study. The present members of the JACC Study and their affiliations are as follows: Dr. A. Tamakoshi (present chairman of the study group) and Dr Y. Ohno (former chairman of the study group), Nagoya University Graduate School of Medicine; Dr. M. Mori, Sapporo Medical University School of Medicine; Dr. Y. Motohashi, Akita University School of Medicine; Dr. I. Tsuji, Tohoku University Graduate School of Medicine; Dr. Y. Nakamura, Jichi Medical School; Dr. H. Iso, Institute of Community Medicine, University of Tsukuba; Dr. H. Mikami, Chiba Cancer Center; Dr. Y. Inaba, Juntendo University School of Medicine; Dr. Y. Hoshiyama, Showa University School of Medicine; Dr. H. Suzuki, Niigata University Graduate School of Medical and Dental Sciences; Dr. H. Shimizu, Gifu University School of Medicine; Dr. H. Toyoshima, Nagoya University Graduate School of Medicine; Dr. S. Tokudome, Nagoya City Graduate School of Medicine; Dr. S. Hashimoto, Fujita Health University School of Medicine; Dr. Y. Ito, Fujita Health University School of Health Sciences; Dr. S. Kikuchi, Aichi Medical University School of Medicine; Dr. A. Koizumi, Graduate School of Medicine, Kyoto University; Dr. T. Kawamura, Kyoto University Center for Student Health; Dr. Y. Watanabe and Dr. T. Miki, Kyoto Prefectural University of Medicine; Dr. C. Date, Mukogawa Women's University; Dr. K. Sakata, Wakayama Medical University; Dr. T. Nose, Tottori University Faculty of Medicine; Dr. N. Hayakawa, Research Institute for Radiation Biology and Medicine, Hiroshima University; Dr. T. Yoshimura, University of Occupational and Environmental Health; Dr. K. Fukuda, Kurume University School of Medicine; Dr. N. Okamoto, Kanagawa Cancer Center; Dr. H. Shio, Shiga Medical Center; Dr. T. Kitagawa, Cancer Institute of the Japanese Foundation for Cancer Research; Dr. T. Kuroki, Gifu University; and Dr. K. Tajima, Aichi Cancer Center Research Institute. The past investigators of the study group are listed in Ref. 11), except for the following seven members (affiliations are those at the time when they participated in the study): Dr. T. Shimamoto, University of Tsukuba; Dr. H. Tanaka, Tokyo Medical and Dental University; Dr. S. Hisamichi, Tohoku University Graduate School of Medicine; Dr. M. Nakao, Kyoto Prefectural University of Medicine; Dr. T. Suzuki, Osaka Medical Center for Cancer and Cardiovascular Diseases; Dr. T. Hashimoto, Wakayama Medical University; and Dr. T. Ishibashi, Asama General Hospital. This work was supported by a Grant-in-Aid for Scientific Research on Priority Areas (2) (No. 14031221) from the Ministry of Education, Culture, Sports, Science and Technology of Japan. The JACC Study has also been supported by Grants-in-Aid for Scientific Research (Nos. 61010076, 62010074, 63010074, 1010068, 2151065, 3151064, 4151063, 5151069, 6279102, and 11181101) from the same Ministry.

1. Hsing AW, Devesa SS. Trends and patterns of prostate cancer: what do they suggest? *Epidemiol Rev* 2001; **23**: 3–13.
2. International Agency for Research on Cancer (IARC). Cancer incidence in five continents. vol VII. Lyon (France): IARC; 1997. p. 66–8, 383–405, 872–3.
3. Barnes S. Role of phytochemicals in prevention and treatment of prostate

cancer. *Epidemiol Rev* 2001; **23**: 102–5.
4. Adlercreutz H. Phyto-oestrogens and cancer. *Lancet Oncol* 2002; **3**: 364–73.
5. Stattin P, Adlercreutz H, Tenkanen L, Jellum E, Lumme S, Hallmans G, Harvei S, Teppo L, Stumpf K, Luostarinen T, Lehtinen M, Dillner J, Hakama M. Circulating enterolactone and prostate cancer risk: a Nordic nested case-control study. *Int J Cancer* 2002; **99**: 124–9.
6. Akaza H, Miyamaga N, Takashima N, Naito S, Hirao Y, Tsukamoto T, Mori M. Is daidzein non-metabolizer a high risk for prostate cancer? A case-controlled study of serum soybean isoflavone concentration. *Jpn J Clin Oncol* 2002; **32**: 296–300.
7. Hsing AW. Hormones and prostate cancer: what's next? *Epidemiol Rev* 2001; **23**: 42–58.
8. Shaneyfelt T, Husein R, Bubley G, Mantzoros CS. Hormonal predictors of prostate cancer: a meta-analysis. *J Clin Oncol* 2000; **18**: 847–53.
9. Ohno Y, Tamakoshi A, the JACC Study Group Japan. Collaborative cohort study for evaluation of cancer risk sponsored by Monbusho (JACC Study). *J Epidemiol* 2001; **11**: 144–50.
10. World Health Organization (WHO). International statistical classification of diseases and related health problems. 10th Rev. vol I. Geneva (Switzerland): WHO; 1992.
11. Coward L, Kirk M, Nicolas A, Stephen B. Analysis of plasma isoflavones by reversed-phase HPLC-multiple reaction ion monitoring-mass spectrometry. *Clin Chim Acta* 1996; **247**: 121–42.
12. Watanabe S, Yamaguchi M, Sobue T, Takahashi T, Miura T, Arai Y, Mazur W, Wähälä K, Adlercreutz H. Pharmacokinetics of soybean isoflavones in plasma, urine, and feces of men after ingestion of 60 g baked soybean powder (kinako). *J Nutr* 1998; **128**: 1710–5.
13. Lampe JW, Skor HE, Li S, Wähälä K, Howald WN, Chu C. Wheat bran and soy protein feeding do not alter urinary excretion of the isoflavan equol in premenopausal women. *J Nutr* 2001; **131**: 740–4.
14. Setchell KDR, Brown NM, Lydeking-Olsen E. The clinical importance of the metabolite equol-a clue to the effectiveness of soy and its isoflavones. *J Nutr* 2002; **132**: 3577–84.
15. Catalona WJ, Richie JP, Ahmann RR, Hudson MA, Scardino PT, Flanigan RC, deKernion JB, Ratliff TL, Kavoussi LR, Dalkin RL, Waters WB, MacFarlane MT, Southwick PC. Comparison of digital rectal examination and serum prostate specific antigen in the early detection of prostate cancer: results of a multicenter clinical trial of 6,630 men. *J Urol* 1994; **151**: 1283–90.
16. Nakagawa S, Watanabe M, Nomoto T, Sugimoto K, Mikami K, Ebisui K, Kitamura K, Kojima M, Saitoh M, Watanabe H. Mass screening for prostate cancer. *Acta Urol Jpn* 1997; **43**: 447–52.
17. Stamey TA, Kabalin JN, McNeal JE, Johnstone IM, Freiha F, Redwine EA, Yang N. Prostate specific antigen in the diagnosis and treatment of adenocarcinoma of the prostate. II. Radical prostatectomy treated patients. *J Urol* 1989; **141**: 1076–83.
18. Partin AW, Carter HB, Chan DW, Epstein JI, Oesterling JE, Rock RC, Weber JP, Walsh PC. Prostate specific antigen in the staging of localized prostate cancer: influence of tumor differentiation, tumor volume and benign hyperplasia. *J Urol* 1990; **143**: 747–52.
19. Shida K, Koyanagi T, Kumamoto Y, Ohori T, Suzuki K, Origas S, Yamanaka H, Koiso K, Shimazaki J, Hosaka M, Sato S, Katayama T, Takeuchi H, Miyake K, Tomoyoshi T, Watanabe H, Goto H, Ohmori H, Tanaka H, Fujita Y, Kumazawa J, Sakamoto K, Saito Y, Ikegami K, Kuriya N. Clinical evaluation of TZP-61 on benign prostatic hypertophy, a double-blind comparative study with Prostal® 25. *Rinsho Iyaku* 1988; **4**: 1145–64 (in Japanese).
20. Itoh N, Kumamoto Y, Akagashi K, Murata H, Tsukamoto T, Umehara T, Mikuma N, Yamaguchi Y, Nanbu A, Suzuki N. The assessment of bioavailable androgen levels from the serum free testosterone level. *Jpn J Endocrinol* 1991; **67**: 23–32.
21. Walsh PC (ed). Endocrine control of prostate growth. In: Campbell's urology. 8th ed. Philadelphia (PA): Saunders; 2002. p. 1245–8.
22. Breslow NE, Day NE. Statistical methods in cancer research. vol I. Lyon (France): IARC; 1980. p. 247–79.
23. Lampe JW, Gstafson DR, Hutchins AM, Martini MC, Li Sue, Wähälä K, Grantis GA, Potter JD, Slavin JL. Urinary isoflavonoid and lignan excretion on a Western diet: relation to soy, vegetable, and fruit intake. *Cancer Epidemiol Biomarkers Prev* 1999; **8**: 699–707.
24. Madersbacher S, Schatzl G, Bieglmayer C, Reiter WJ, Gassner C, Berger P, Zidek T, Marberger M. Impact of radical prostetectomy and TURP on the hypothalamic-gonadal hormone axis. *Urology* 2002; **60**: 869–74.
25. Nagata C, Inaba S, Kawakami N, Kakizoe T, Shimizu H. Inverse association of soy product intake with serum androgen and estrogen concentrations in Japanese men. *Nutr Cancer* 2000; **36**: 14–8.
26. Nagata C, Takatsuka N, Shimizu H, Hayashi H, Akamatsu T, Murase K. Effect of soymilk consumption on serum estrogen and androgen concentration in Japanese men. *Cancer Epidemiol Biomarkers Prev* 2001; **10**: 179–84.
27. Coldham NG, Dave M, Sivapathasundaram S, McDonnell DP, Conner C, Sauer MJ. Evaluation of recombinant yeast cell estrogen screening assay. *Environ Health Perspect* 1997; **105**: 734–42.
28. Breinholt V, Larsen JC. Detection of weak estrogenic flavonoids using a recombinant yeast strain and a modified MCF7 cellproliferation assay. *Chem Res Toxicol* 1998; **11**: 622–9.
29. Morito K, Morito K, Hirose T, Kinjo J, Hirakawa T, Okawa M, Nohara T, Ogawa S, Inoue S, Muramatsu M, Masamune Y. Interaction of phytoestrogens with estrogen receptors alpha and beta. *Biol Pharm Bull* 2001; **24**: 351–6.
30. Markiewicz L, Gary J, Adlercreutz, Grpide E. *In vitro* bioassays of non-steroidal phytoestrogens. *J Steroid Biochem Mol Biol* 1993; **45**: 399–405.
31. Hedlund TE, Johannes WU, Miller GJ. Soy isoflavonoid equol modulates the growth of benign and malignant prostatic epithelial cells *in vitro*. *Prostate* 2003; **54**: 68–78.
32. Duncan AM, Merz-Demlow BE, Xu X, Phipps WR, Kurzer MS. Premenopausal equol excretors show plasma hormone profiles associated with lowered risk of breast cancer. *Cancer Epidemiol Biomarkers Prev* 2000; **9**: 581–6.
33. Izumi T, Piskula MK, Osawa S, Obata A, Tobe K, Saito M, Kataoka S, Kubota Y, Kikuchi M. Soy isoflavone aglycones are absorbed faster and in higher amounts than their glucosides in humans *J Nutr* 2000; **130**: 1695–9.
34. Yamamoto S, Sobue T, Sasaki S, Kobayashi M, Arai Y, Uehara M, Adlercreutz H, Watanabe S, Takahashi T, Iitoi Y, Iwase Y, Akabane M, Tsugane S. Validity and reproducibility of a self-administered food-frequency questionnaire to assess isoflavone intake in a Japanese population in comparison with dietary records and blood and urine isoflavones. *J Nutr* 2001; **131**: 2741–7.
35. Morton MS, Arisaka O, Miyake N, Morgan LD, Evans BA. Phytoestrogen concentrations in serum from Japanese men and women over forty years of age. *J Nutr* 2002; **132**: 3168–71.